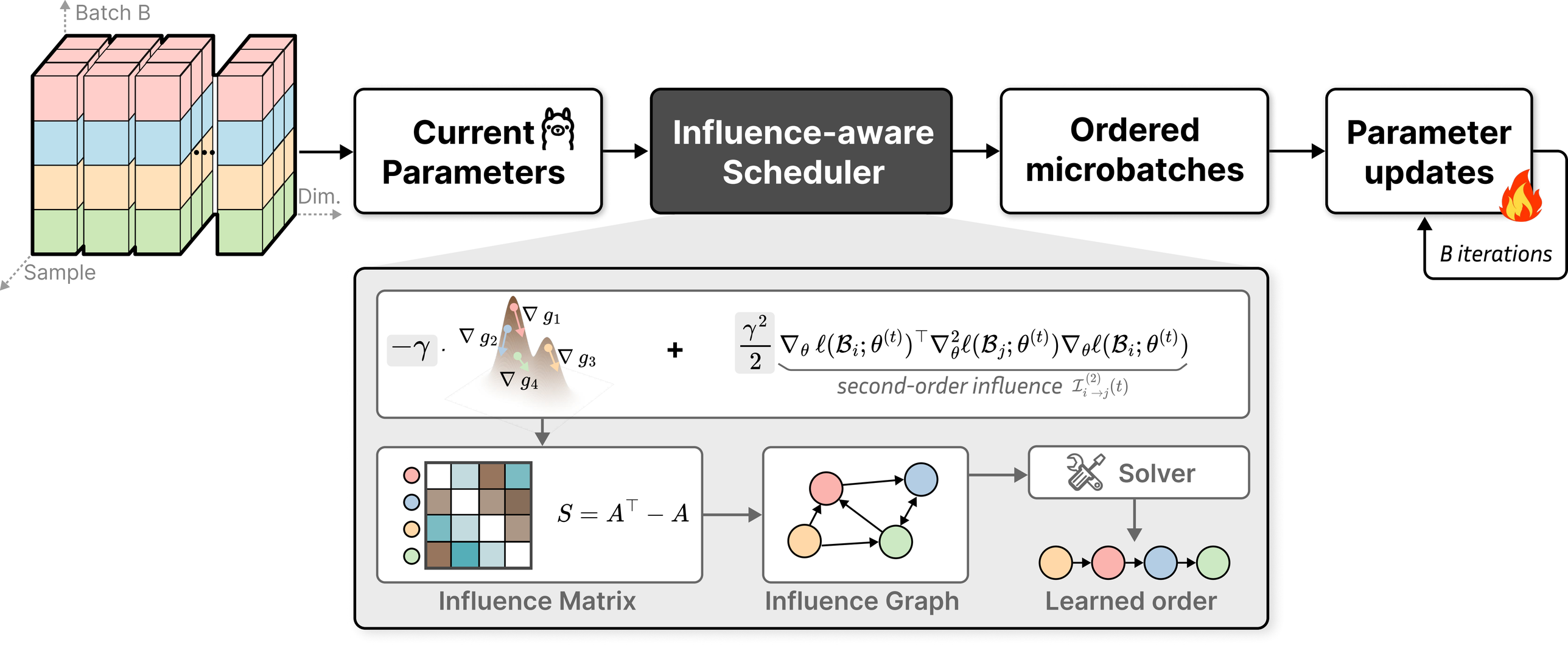
Batch B
Dim.
Sample
Current Parameters
Influence-aware Scheduler
Ordered microbatches
Parameter updates
B iterations
$-\gamma \cdot \nabla g_1 \; \nabla g_2 \; \nabla g_3 \; \nabla g_4$
+
$\frac{\gamma^2}{2} \underbrace{\nabla_\theta \ell(\mathcal{B}_i;\theta^{(t)})^\top \nabla^2_\theta \ell(\mathcal{B}_j;\theta^{(t)}) \nabla_\theta \ell(\mathcal{B}_i;\theta^{(t)})}_{\text{second-order influence } \mathcal{I}^{(2)}_{i\to j}(t)}$
$S = A^\top - A$
Influence Matrix
Influence Graph
Solver
Learned order